



Subham Sharma
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23M1838
M.Tech.
Gender: Male
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Examination	University	Institute	Year	CPI / %
Post Graduation	IIT Bombay	IIT Bombay	2025	8.59
Graduation	JNVU Jodhpur	MBM Jodhpur	2020	6.72
Graduation Specialization: Production & Industrial Engineering				

PROJECTS

Heat and Mass Balance in Electric Arc Furnace | MTech Thesis

(May'24-present)

- Optimization of Lime, Scrap, Hot-Metal, electric power and oxygen by using **Hot DRI** (approx. 33%) in charge mix
- Performed mass and heat balance calculations based on industrial data to **optimize** processes using Python, SciPy, Matplotlib etc.
- Utilized **modelling techniques** to accurately **predict carbon** and oxygen content in steel at the conclusion of the refining process, enhancing process control and **quality** assurance

EyeView- real time video conferencing platform | Self Project

(July'24-Sep'24)

- Built a Zoom-like video conferencing platform with **MERN stack**, integrating real-time audio, video, and chat using **WebRTC** and **Socket.IO**.
- Implemented secure authentication and session management with **JWT** and **Passport.js**, ensuring privacy and collaboration.
- Designed a responsive, user-friendly interface with **React** and Material-UI, leveraging **MongoDB** and **Express.js** for scalable data handling.

Airbnb | Self Project

(Sep'24-Dec'24)

- Added a real-time **Weather feature** to deliver live updates, improving the overall user experience.
- Created a secure user authentication system with **Passport.js** for managing profiles, including **signup** and login functionalities.
- Developed a scalable backend using **Express.js** and **MongoDB**, ensuring robust data management and performance.
- Designed a responsive frontend with **EJS** and **Bootstrap**, focusing on usability and seamless user interaction.

Colosseum | Self Project

(Feb'24-Apr'24)

- Designed the frontend using **React** and **Tailwind**, backend with **express.js**, **Node.js** and managed the database using **MongoDB**
- Implemented a highly **secure** user **login** system using **JWT** and **OAuth 2.0**, ensuring robust authentication and seamless user experience
- Developed **microservices** with **APIs** for recommendations, search, and user profiles, showcasing expertise in service-oriented design

File Hider | Self Project

(Apr'24-June'24)

- Developed a robust file hider application in **Java**, incorporating advanced **user authentication** through **OTP** and **email verification**
- Integrated **MySQL** for secure storage of user credentials and **data encryption** to protect sensitive files from unauthorized access

SOFTWARE SKILLS/KEY COURSES

Software: Node.js | MongoDB | Git | **Programming languages:** Java | HTML | CSS | SQL | JavaScript | **Framework/Libraries:** React | Express.js | Mongoose | Bootstrap | Tailwind CSS | **Courses:** Statistics and Probability for Materials Engineers | IIT Bombay | Java DSA & Full Stack Development | Apna College

POSITIONS OF RESPONSIBILITY

Interview Companion | Institute Placement Team | IIT Bombay

(Oct'23 - Dec'23)

- Coordination efforts with a team of **250+** members to successfully organize and **execute interviews** for more than **2,300** students
- Assisted in **administering tests** for **20+** firms and efficiently managed **student inquiries**, ensuring a smooth and effective process

Student coordinator | Institute student companion programme | IIT Bombay

(Jun'24 - present)

- Selected as a student companion from the **300+** applicants, Providing counsel to 6-students throughout the year, aiding resolution of their academic and non-academic issues
- Part of a team of **230+** coordinators, facilitating smooth transition and holistic development for 1st year Post graduate students

Teaching Assistant | AI and Data Science Lab | IIT Bombay

(Aug'23 - Dec'23)

- Actively mentored & engaged with a group of **280+ UG** students in the AI & DS Lab, providing guidance and support in **programming**
- Assisted the professor in **evaluating assignments** and answer **sheets**, and conducted **crib** sessions to clear doubts of students

SCHOLASTIC ACHIEVEMENTS

- Awarded **AA grade** in **4 courses** and achieved above 9 SGPA in last 2 semesters with exceptional performance in academics (2023-24)
- Secured an impressive **97.27** percentile in **GATE Mechanical 2023** examination amongst **63,000+** candidates who appeared (2023)
- Achieved **96.01** percentile in the **GATE XE 2023** examination, standing out among **11,000+** participated candidates (2023)

EXTRACURRICULAR ACTIVITIES

- Participated in a group dance in **Informal 1, Phase 1** and **Salsa Bollywood Showcase** | IIT Bombay (Aug'23 - Dec'23)
- Accepted challenge of Push India Push **Push-Up Challenge** | FITISTAN | PUSH INDIA PUSH | IIT Bombay (Sep'23)

HOBBIES: Cricket | Dance | Travelling | Gym